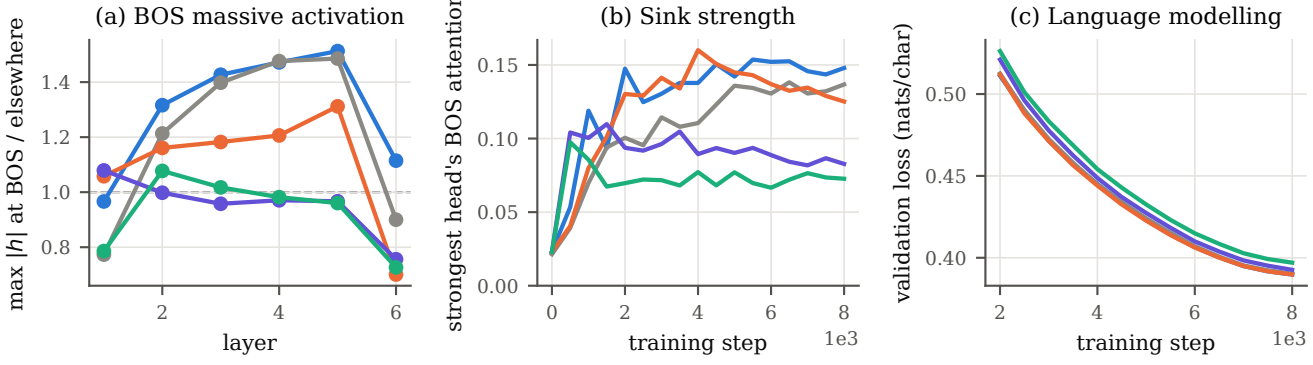
(a) BOS massive activation
max |h| at BOS / elsewhere
1.4
1.2
1.0
0.8
2
4
6
layer
(b) Sink strength
strongest head's BOS attention
0.15
0.10
0.05
0.00
0
2
4
6
8
training step
1e3
(c) Language modelling
validation loss (nats/char)
0.50
0.45
0.40
2
4
6
8
training step
1e3
softmax
off-by-one
linear gate, λ=1
linear gate, λ=10
race-matched gate